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Fast and Seamless Merging of Post-Stack Seismic Volumes

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During exploration, it is common to have access to multiple 3D post-stack seismic volumes, with different acquisition parameters, geometries and data quality that means conducting integrated interpretation and analyses of data between and across surveys can be challenging. Merging a 3D seismic survey would ideally require full re-processing using pre-stack data. However, given time, budgetary, data availability, or compute constraints, this is not always a practical solution. An alternative solution is to merge post-stack seismic volumes, which can provide a suitable substitute for many regional structural interpretation workflows. However, achieving a seamless post-stack seismic merge is challenging and often matching amplitudes, phase, and time shifts consistently cannot be done using bulk corrections across an entire survey due to lateral changes in geology. Seams at the boundaries between seismic surveys or changes in waveform characteristics will create challenges for interpretation, especially for the predictive tools being adopted for automated interpretation workflows. For example, a seam between two datasets is a large discontinuity that will lead to a potential false positive fault prediction. The seam may also prevent horizon interpretation tools from traversing between the two volumes. To reduce these challenges, we present an automated workflow for seamlessly merging two or more post-stack 3D seismic volumes that will enable easier interpretation workflows across regions where interpretations need to span multiple volumes.

Traditionally, survey merging is achieved by applying seismic data interpolation/regularization, phase rotation, amplitude matches, and the optimal bulk time shift obtained to several seismic surveys to condition the different vintages and produce a seamless merged data set. This has been mainly done by designing a post-migration match filter which estimates wavelets in each time zone after isolating trace segments in the defined window and derived by least-squares time-domain with a smoothness constraint. However, after applying those estimated time, phase and amplitude adjustments, the surveys especially their spectra are matched in a loose sense. In case where the low frequency legacy data and high-resolution modern data are required to merge, the final spectrum is closer to low frequency spectrum of legacy data.

The varying levels of resolution, acquisition patterns, and data quality in seismic surveys make their integration complex task. To overcome these challenges a mixture of selected deep-learning, optimization and classical machine-learning algorithms will be utilized to analyze and process data from multiple surveys. These algorithms will identify overlaps in the data and employ advanced techniques to match the spectra and phase of the seismic data, resulting in a seamless transition between different seismic surveys. Ultimately

a complete merged 3D regional seismic survey will be generated. Various machine learning algorithms, including different types of neural networks will be utilized to analyze the data, detect gaps, and fill them with accurate data values.

Introduction

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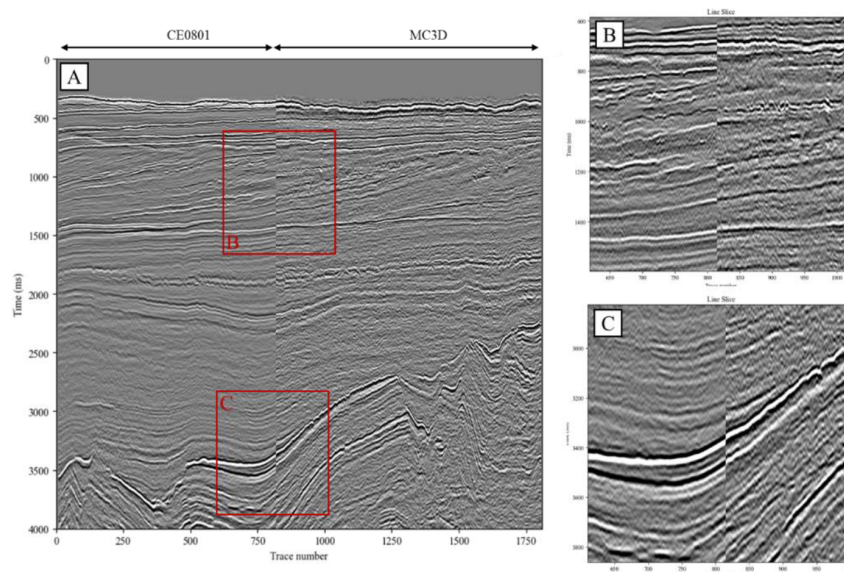


Figure 1—Example of a naïve survey merge between CE0801 and MC3D seismic volume from the Norwegian shelf (NOD, 2023). A) demonstrates a full line merge where the sharp transition between the datasets with clear breaks in the reflectors between the two surveys causing a vertical seam. B) and C) show zoomed in versions of the corresponding red boxes in A).

Method

The proposed automated method for merging starts with the traditional merging techniques of computing bulk corrections for amplitude, frequency, static time and phase shifts to match all input volumes to a single reference survey. For time and phase corrections, a least squared solution is used to solve for the optimal shifts to apply to each survey (Bishop and Nunns, 1994). However, due to lateral variations in geology, these bulk corrections cannot account for all differences between seismic volumes and time shifts may still be observed between the same reflector in different volumes. To further account for lateral changes, we solve a graph based global optimization to obtain a set of spatially varying shifts that will further align reflections ahead of any final merge.

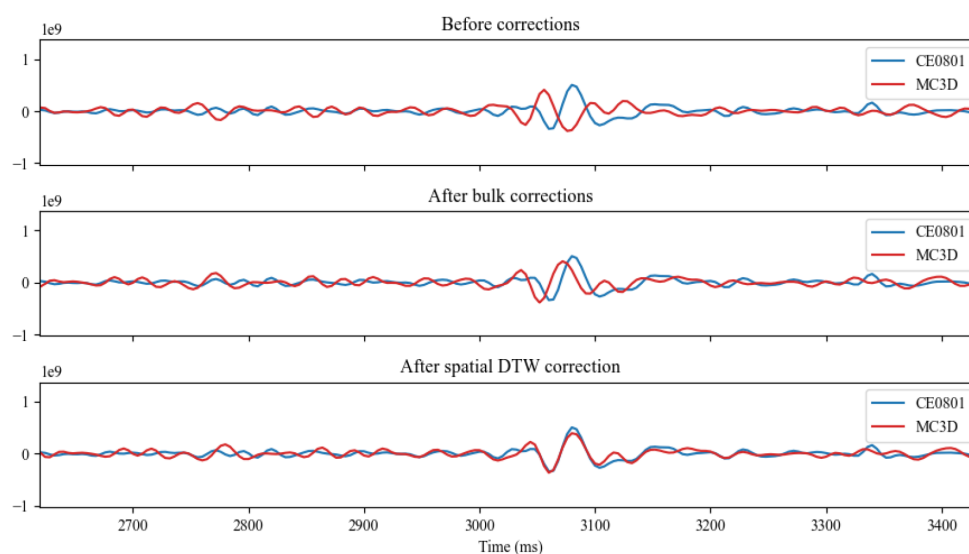


Figure 2—Example of traces with corrections applied to two co-located traces from CE0801 and MC3D seismic surveys on the Norwegian shelf. (NOD, 2023). a) after only amplitude correction is applied b) after bulk corrections applied to correct for phase and static time shift c) after additional DTW corrections are applied.

A graph is initially setup where traces in surveys correspond to nodes and the edges link neighbouring traces within each survey. Additional edges are also added to connect approximately co-located traces in different surveys. The result is a graph with a single connected component. Where traces connected by an edge are located in different surveys, a modified version of the Dynamic Time Warping for seismic traces (DTW; Hale, 2013) is used to further refine trace alignment. In our modified DTW, the alignment error is computed on windowed cross-correlation envelopes as opposed to the root mean square error between samples. This allows stable shifts to be computed between surveys, which align reflectors, but are phase agnostic. To further ensure stability, the following global linear system is constructed from the graph, where the solved shifts (s) will align reflections consistently across all input surveys.

$$\begin{bmatrix} w(s(i, z) - s(i+1, z + d_{i,z})) \\ \mu \frac{\partial s}{\partial z} \\ \mu \frac{\partial s}{\partial h} \end{bmatrix} \approx \begin{bmatrix} wd(i, z) \\ 0 \\ 0 \end{bmatrix} \quad \text{Equation 1}$$

The first line encodes the inter-survey trace pair shifts computed by DTW with the left hand stating that the difference between final shifts on one trace of a co-located trace pair at a specified z -value (time or depth) $s(i, z)$, and the final shift values on the corresponding trace at a z -value shifted by the DTW result for that trace pair, $s(i, z + d_{i,z})$, should be equal to the DTW result, $d_{i,z}$. The weighting term, w , allows this term to be weighted by the quality of the DTW alignment score. The second and third lines of the equation are smoothing constraints for the final shifts. They weakly constrain the vertical, $\frac{\partial s}{\partial z}$, and horizontal, $\frac{\partial s}{\partial h}$, gradients of the output shifts, s , to be equal to zero, thus preventing the output shifts from changing too rapidly and smoothing the overall output. Smooth shifts are critical to ensure the output seismic is not locally distorted, for example by misalignment of an individual trace-pair during DTW.

To ensure datasets are merged seamlessly, it is necessary to calculate blending masks for each volume. These masks determine how to weight traces from different volumes in the final merge such that the data seamlessly transitions from one dataset to the next.

To do this, a binary segmentation mask is first created, which defines the portion of the seismic volume where good quality seismic data exists. The second step creates a smooth weighting mask that is defined based on the Euclidean distance of any sample to the nearest data edge in the segmentation mask. The result of this is that a point receives a higher weighting when it lies near the center of a seismic volume, away from the edges where there may be lower quality data or imaging artefacts.

The final merged volume can be created by applying all bulk corrections including amplitude, frequency, static time and phase shifts as well as the spatially varied global time shifts, to the input volumes. Secondly, corrected data is interpolated from the input seismic grids onto the output seismic grid. Final stacking of overlapping traces in the merged volume is done according to the weighted blending mask.

a. Trace Balancing

The primary goal of the trace balancing section of the workflow is to align the amplitudes, phase and frequency of the input surveys to be merged. It is common that due to different acquisition and processing histories datasets acquired in the same region may have significantly different characteristics, for example amplitudes may not be scaled the same, the phase of each dataset may be different. Without matching the waveform characteristics between each survey, merging post-stack seismic datasets will lead to unwanted artifacts, such as vertical discontinuities or amplitude reductions in the final merged seismic image. This section describes how the SSM tool can be used to automatically balance the amplitude, phase and frequency of input seismic volumes as the first step of the merging workflow.

i. Amplitude

To balance seismic amplitudes a simple strategy is used, with the aim of matching the mean and standard deviation of the seismic data from each volume. The mean and standard deviation of each survey is calculated efficiently by randomly sampling N seismic traces from each volume. For the ADNOC seismic volumes and for this POC we used 20,000 traces to gain statistically meaningly full values without having to compute over the entire volume. Once the mean and standard deviation is calculated for each dataset a mapping from one survey to another can be calculated as follows:

$$A_{2bal} = m_1 + (A_2 - m_2) \times \frac{s_1}{s_2} \quad \text{Equation 3.1}$$

Where the mean of the input survey (m_2) is subtracted from its amplitude values (A_2), with the mean of the target survey (m_1) the added. Finally, these amplitude values are multiplied by the ratio of the target survey standard deviation (s_1) over the standard deviation for the input survey (s_2). The amplitudes of the input survey (A_{2bal}) will now have the same mean and standard deviation as the target survey.

ii. Phase

To balance seismic phase between two datasets, an optimal phase rotation between the two datasets needs to be determined. The SSM tool does this by analyzing N co-located trace pairs in the overlapped region of the seismic volumes to be merged. For each trace pair, the input survey trace is phase rotated through 180° , at 1° intervals a correlation score is determined and added to the list of results. After all N trace pairs have been processed, the optimal phase rotation to align the two surveys is chosen as the angle that resulted in highest correlation score across all trace pairs.

Due to differing processing velocity models and processing histories, it is possible that co-located traces from different surveys cannot be aligned via a phase rotation since there may also exist a vertical shift in the seismic trace. To therefore determine if a phase rotation needs to be applied, the vertical shift must first be corrected for. The process for doing this is outline in greater detail in section d.ii but uses Dynamic Time Warping (DTW) on the amplitude envelope of two traces to pre-align them prior to calculating the optimal phase rotation. The amplitude envelope is used as it is independent of phase so traces can be aligned, even if they are of different phase.

iii. Frequency

The frequency content of different seismic volumes can be substantially different when different acquisition parameters or processing workflows are applied to the data. A component of blending two seismic volumes therefore matching the power spectra of the two seismic volumes as best possible.

A simple way to achieve this is by applying a function that normalizes the power spectra of the two datasets. The first step in the workflow is therefore computing the average power spectra of each seismic volume to be merged using N sampled traces from each volume. From these power spectra a filter can be designed that will smoothly balance the power spectra of the two surveys. If we want to match the spectra of survey 1 to survey 2 then the filter is designed as follows

$$filter = \left\langle \frac{power_1}{power_2} \right\rangle \quad \text{Equation 3.2}$$

Where $\langle . \rangle$ is a smoothing function designed to ensure the filter is smooth. The filter is then applied by multiplying with a trace in the frequency domain. An example of this filter design is shown in Figure. This is similar to the idea of spectral whitening (Yilmaz, 2001), only we don't whiten the spectrum uniformly. The filter can also be modified with high and low cutoffs, setting values for frequencies outside the range we wish to modify to 1. This can help to ensure we don't amplify noise when applying the filter, particularly in the high frequencies.

b. Geometry Optimization to match data in time or depth

Despite matching amplitude, phase and frequency content of different seismic datasets. These corrections do not account for potential time or depth shifts that may exist between datasets. For the purpose of this report the workflow will be discussed with respect to time shifts, since ADNOC data was provided in the time domain. However, the same methodology could be applied to data in the depth domain, where this step in the workflow is even more crucial due depths being highly dependent on the seismic velocity model which may be different between different datasets.

To accurately and efficiently shift seismic volume to match in time a two-step strategy is used by the SSM tool. The first step utilizes Dynamic Time Warping (DTW) on seismic trace pairs that are co-located in each dataset. The second step used the shifts calculated in the overlap region in the first step and performs a global optimization to compute the optimal shifts that will shift both the entirety of the seismic volumes so that they align in time.

i. Graph generation

In order to merge seismic surveys SSM utilizes an underlying graph data structure to carry out much of the processing from here on out. The graph makes it easy to identify co-located seismic traces in the overlap region between seismic datasets and also to perform the global shift optimization (section iii) which is crucial to getting an accurate alignment prior to merging the seismic volumes.

To generate the graph data structure traces are added to the graph as nodes and intra-survey edges are added between adjacent traces on the seismic grid. To connect two independent seismic surveys inter-survey edges are added between traces (nodes) that are co-located within a set tolerance. It is these inter-survey edges that connect co-located traces from different surveys that create a single connected component in the graph that is used to solve global shifts later on.

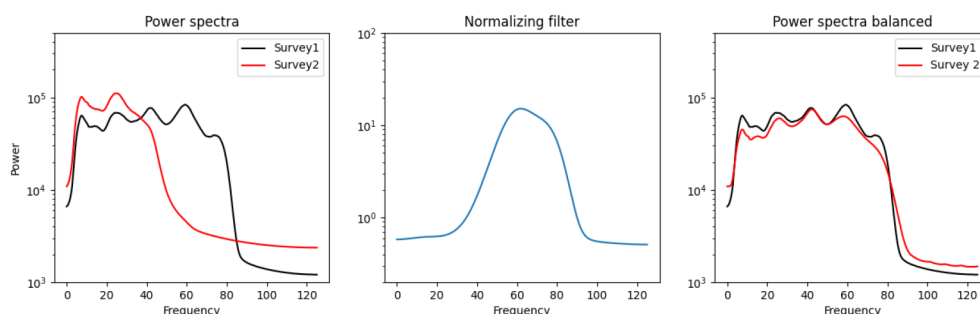
ii. 1D trace shifts

All co-located trace pairs from different surveys defined by the graph are used to compute shifts via Dynamic Time Warping (DTW) that will correct for any misalignment of the traces in depth. DTW in its simplest form searches for an optimal path through an alignment error matrix that will shift or "warp" the traces so that they align (Sakoe & Chiba, 1978). Seismic data is, however, inherently cyclical which can make it relatively unstable and prone to cycle skipping if used conventionally. To overcome this SSM replaces the alignment error matrix, which is typically based on the root mean squared error (RMSE) between trace samples, with an alignment error matrix based on cross-correlation windows. This helps to make the final shifts much more stable. Another strategy used by SSM to ensure accurate and stable shifts is the use of "strain" in the path finding part of the algorithm (Hale, 2013). Strain helps again to ensure a smooth "path" is found that aligns the seismic traces and prevents shifts from varying too quickly which could distort the final seismic data.

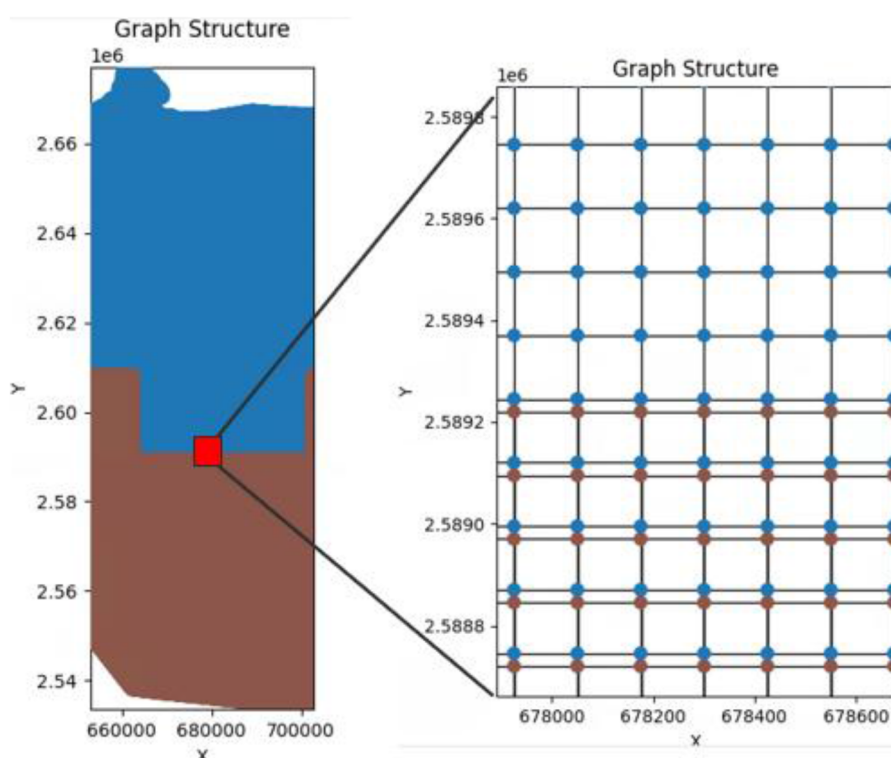
iii. 3D global shift optimization

The previous step of the workflow computes shifts that will align traces in the overlap region between two seismic volumes. The next step is to compute shifts across the entirety of the volumes to be merged so that they can both be shifted in order to align reflections. This is done in SSM using a solving a global optimization problem across the surveys to be merged as shown below.

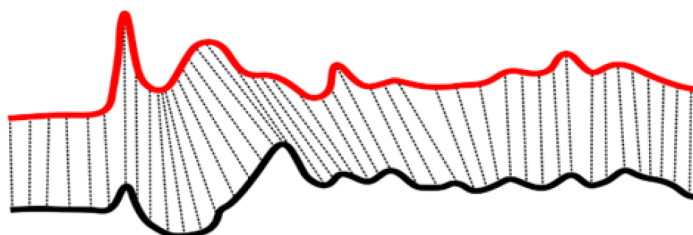
$$\begin{bmatrix} w(s(i, z) - s(i+1, z + d_{iz})) \\ \mu \frac{\delta s}{\delta z} \\ \mu \frac{\delta s}{\delta z} \end{bmatrix} \approx \begin{bmatrix} wd(i, z) \\ 0 \\ 0 \end{bmatrix} \quad \text{Equation 3.3}$$



Figure—Arbitrary example of designing a filter to balance the mean power spectra of two surveys.
The left-hand plot shows the original mean power spectra for two arbitrary surveys. Center plot shows the filter designed to balance the two power spectra, which is $(\text{power1}/\text{power2})$ with additional smoothing applied. The right-hand plot shows the recomputed mean power spectra for survey 1 and 2 where the normalizing filter was applied to traces in survey 2 to match the power spectra of survey 1.



Figure—Example of graph generated for cropped regions of the overlapping onshore block 1 (blue) and onshore block 2 (brown) seismic datasets. Left is the overall graph, and right is a zoomed in region highlighted by the red box. In the zoomed in plot, black inter-survey edges connect adjacent nodes (traces) in the same dataset. Any two approximately co-located blue and brown nodes (traces) are also connected by an additional edge which links the two surveys together in the graph.



Figure—Schematic diagram of Dynamic Time Warping algorithm and how it computes shifts to align two signals.

The first line of the equation to be solve encodes the trace pairs shifts that were calculated in the previous step using DTW. The left-hand side states that the difference in the final shifts between co-located traces

connected by an inter-survey edge in the graph should equal the right-hand side, which is the corresponding shifts computed by DTW for the same trace pair. Both sides contain a weighting term to preferentially use shifts that have a higher alignment score from DTW. The second and third lines of the equation are smoothing terms that control the lateral and vertical extent of smoothing in the final shifts which is controlled by the parameter μ . The smoothing is important as it ensures final shifts transition smoothly towards a background shift value as you move away from the overlap region of the two datasets. Without this we might risk introducing false structures into the seismic data which is something that must be avoided.

Solving Equation 3.2 for the shifts (s) will provide shifts to align the surveys in time. When solving it is typical to keep one survey fixed as the reference and all shifts are computed to align data to the reference survey. This is what has been implemented in the current POC, Onshore Block 1 was chosen as the reference survey and shifts were solved to align Onshore Block 2 to Block 1. However, it should be noted it is also possible to not use a reference survey and, in this case, shifts will be averaged across both surveys in order to align them.

Workflow

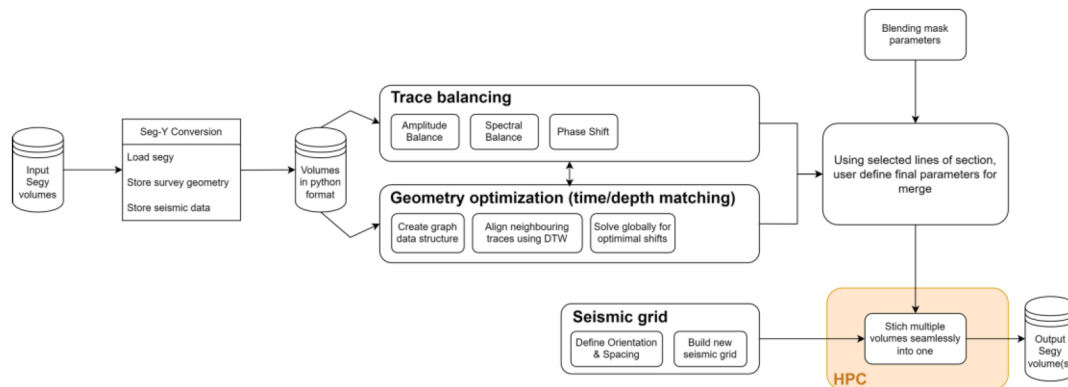


Figure 3.1—Workflow for seismic survey merging technology

Results

Here we present results from the merge of two 3D stacked seismic volumes, taken from the CE0801 and MC3D seismic surveys on the Norwegian shelf (NOD, 2023). In this example, CE0801 was chosen as the reference survey, to which the MC3D is aligned. Figure 2 shows the corrections applied to a single co-located trace between the two surveys, it shows that bulk corrections, consisting of a 0.5 ms time shift and -143° phase rotation, can approximately align waveforms from MC3D to CE0801. However, the addition of shifts computed from DTW further refine the alignment of the two waveforms, which will result in better stacking of the two surveys in the overlap region of the final survey merge.

Figure 3 shows a slice through the final merged 3D volume created from the original CE0801 seismic volumes and the corrected MC3D seismic volumes and the smooth weighted blending mask. The final result demonstrates excellent continuity of reflectors and preservation of amplitudes in the transition region between the two datasets, suggesting waveforms were correctly aligned prior to the merge. The result makes for easier geological interpretation of horizon and faults, particularly across the transition region between the two datasets.

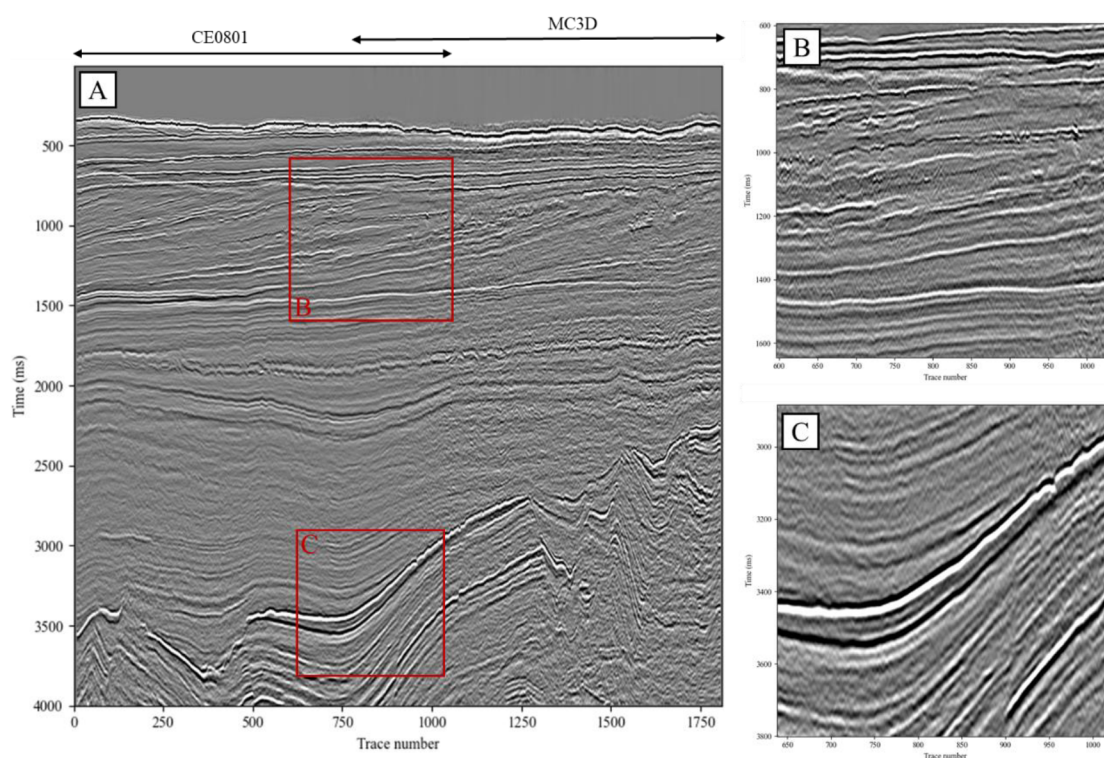


Figure 3—Example of a seamless merge between CE0801 and MC3D seismic volume from the Norwegian shelf (NOD, 2023). A) demonstrates a full line merge where, due to correct amplitudes, phase, and spatially variable time shifts a seamless transition is observed between the two surveys with excellent lateral continuity of reflectors. B) and C) show zoomed in versions of the corresponding red boxes in A).

Conclusions

An automated method is proposed for merging two or more seismic volumes to generate a single seamless seismic volume with consistent amplitudes, phase, frequency and alignment of reflectors. The method combines traditional geophysical techniques for seismic mis-tie analysis such as bulk amplitude, phase, and time-shift, however, it also uses spatially variable and optimized time-shifts to further refine the alignment of these two surveys. The result is a merging procedure that can be applied in hours rather than months. This enables merged volumes to easily form the starting point for large regional studies that span multiple seismic surveys and aim to utilize modern predictive tools that might struggle to predict across transition regions between surveys that have been poorly merged.

Acknowledgements

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References

- Bishop, T.N. and Nunns, A.G., 1994. Correcting amplitude, time, and phase mis-ties in seismic data. *Geophysics*, **59**(6), pp.946–953.
- Hale, D., 2013. Dynamic warping of seismic images. *Geophysics*, **78**(2), pp.S105–S115.
- The Norwegian Offshore Directorate (NOD), 2023. Available at: <https://www.sodir.no/en/diskos/seismic/> (Accessed: 20 November 2024).